Decodability Does Not Establish Predictive Privilege: Testing Activation Monitoring Across Observation Boundaries and Readouts

Anonymous Author(s)
Affiliation
Address
email

Abstract

White-box monitoring studies sometimes compare a post-deliberation activation with text observed before deliberation, so representation type and observation timing change together. We isolate this confound in a binary Diplomacy task. Post-deliberation activations outperform pre-deliberation text (Brier 0.103 vs. 0.231); when the text observer receives the deliberation that precedes that activation, its Brier falls to 0.033. A repeated-trajectory follow-up tests action-specific variation within the same strategic states. Late activations remain extremely predictive, while direct same-family target scoring brings text ranking close to activation (AUROC 0.995 vs. 1.0). We use *predictive privilege* for the stronger claim that an activation advantage belongs to the representation rather than to the evidence boundary or readout. Across both experiments, the apparent activation advantage is conditional on when each channel is observed and how its evidence is extracted.

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1 Introduction

White-box monitors can inspect internal model states that ordinary text interfaces never expose. Recent work has decoded latent truth, confidence, safety-relevant labels, and future behaviour from language-model activations [3, 2, 13, 4]; some studies report better monitoring performance from activations than from text-only baselines [15, 14]. If an activation is recorded after deliberation D while the text baseline sees only the preceding state S, the comparison changes both the channel and the amount of trajectory observed.

Let S be a decision state, D a generated deliberation, and Y the subsequent action. Our fixed-action experiment records a residual-stream activation $h(S, D)$ and asks a text observer to predict the same Y either from S alone or from (S, D). The activation record, realised action, and generated deliberation stay fixed. Only the visible evidence changes.

Diplomacy is convenient for this design because the action can be checked mechanically. We use one movement-phase state from each of 40 released CICERO games, choose two legal focal orders in advance, and hold the focal power's remaining orders fixed [12]. A local language-model subject deliberates over the pair and submits one order. The source game is kept as the independent sampling unit, including in later experiments that draw several trajectories from the same position.

The fixed-action comparison produces a clear reversal in activation-versus-text ordering. Post-deliberation activation reaches 0.103 Brier; text is at 0.231 before the deliberation and 0.033 after it. We then recapture both channels at five matched observation boundaries. Separately, an exposure/redaction series varies the text available to observers; shuffled-label and model-family checks probe the robustness of the comparison.

One label per state cannot distinguish state-level propensity from trajectory-specific information, so we resample each state eight times. The late residual-stream activation becomes extremely predictive under complete-state holdout; that pattern could reflect an earlier commitment, output preparation, or both. Holding the trajectories fixed, we then change the text readout and obtain substantially different scores. We use *decodability* in a narrow sense: a specified reader can recover the target from a representation. *Accessibility* concerns how readily that target can be extracted within a specified class of plausible readers or readout constraints. *Predictive privilege* is the stronger comparative claim that an activation advantage belongs to the representation itself rather than to later evidence or a weak comparator. Accessibility can itself be the source of such an advantage, so the comparison must specify the reader regime as well as the observation boundary. Predictive privilege is a performance claim, not the still stronger claim that target-relevant information is literally absent from text. Both experiments predict the focal action while probing timing and readout in different ways.

The empirical results are organised by role. Sections 4.1–4.2 establish the primary observation-timing result, first at the fixed-action endpoints and then across matched deliberation boundaries. Section 4.3 collects robustness and extension checks that are not needed to follow the main argument. Section 5 is a separate repeated-trajectory follow-up that reuses the same source states to distinguish trajectory-specific action information from state-level propensity and to test sensitivity to text readout.

2 Related work

Activation monitors increasingly predict later model behaviour. Ashok and May do so before output begins, while Moreno Cencerrado et al. probe pre-generation activations for future answer correctness [2, 13]. Patel and Wang compare activation and text monitors under adversarial pressure [15]. Chan et al. provide the closest temporal comparison, using partial chain-of-thought text or activations captured at the same truncated reasoning point [4]. They generate a new response from each truncation; ours keeps the realised action and deliberation fixed while varying only the text evidence.

Comparative work also makes the baseline part of the claim. Parrack et al. measure deception-probe gains from black-box to white-box monitoring [14]. Song et al. define privileged self-access relative to comparable third-party recovery, while Singh et al. show that apparent introspective advantages can disappear against input-only controls [18, 17]. Guan et al. find that additional test-time compute can improve chain-of-thought monitorability [6]. Reader capacity is therefore part of what a channel comparison establishes.

Probe performance raises a separate interpretive issue. Control tasks test whether success reflects the representation rather than classifier flexibility [7]; amnesic probing removes decoded information and tests the resulting behaviour [5]. Visible chain-of-thought also need not faithfully describe the computation producing an action [10, 9]. Our permutation control repeats layer and regularisation search under shuffled labels, including reader selection in the null. We treat decoding as evidence of recoverability rather than causal use. The experiments below keep the prediction task constant while varying the text channel along two axes: when it is observed and how it is read.

3 Experimental framework

3.1 Decision task and sampling unit

The evaluation set contains 40 movement-phase states from 40 distinct released CICERO games. A Diplomacy movement-phase action contains one order for each controlled unit; in each selected position we vary one focal order between two legal candidates, denoted A and B, while holding the focal power's remaining orders fixed. We refer to those fixed remaining orders as the non-focal orders. State selection and the two candidates are fixed before running the local language-model subject. The model-visible input contains the board state, messages, the non-focal orders, and the two focal candidates. Every monitor therefore faces the same binary prediction target, and the submitted order can be replayed from the released parent state.

One game contributes one independent strategic situation. The fixed-action experiment has one realised outcome per game. The repeated-trajectory follow-up later draws eight trajectories from each of the same states; those draws are repeated observations within a strategic situation, so splits and uncertainty calculations remain grouped by source state.

The main subject, Qwen2.5-7B-Instruct [16], produces a generated deliberation about the two focal orders and then emits a delimited final choice; both candidates appear in the prompt. The 7B cross-family observer and subject replication use Mistral-7B-Instruct-v0.3 [8], while the 32B family/scale extension uses Qwen3-32B [21] and OLMo-2-0325-32B [19].

3.2 Text exposure and activation capture

We use *reader* for any mapping from a channel to a prediction, *observer* for a prompted language-model reader, and *readout* for the extraction procedure. The activation reader is the fitted logistic classifier over residual-stream features. Figure 1 contrasts the fixed-action comparison developed here with the repeated-trajectory follow-up in Section 3.4.

For panel (a), let L_A denote held-out Brier loss for an activation reader recorded after deliberation. A text observer evaluated before deliberation has loss L_{T_0}; the same prompted observer architecture evaluated after receiving the exact deliberation has loss L_{T_1}. We report

$$\Delta_0 = L_{T_0} - L_A, \tag{1}$$

$$\Delta_1 = L_{T_1} - L_A, \tag{2}$$

where positive values favour activation. Their difference is

$$\Delta_0 - \Delta_1 = L_{T_0} - L_{T_1}. \tag{3}$$

The difference between the two contrasts is exactly the observer's gain from seeing the deliberation.

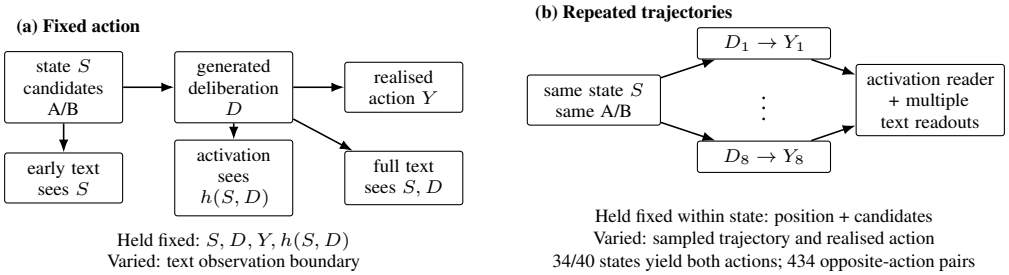


Figure 1: **The two experiments isolate different sources of apparent monitoring advantage.** (a) Fixed action holds the stored trajectory and post-deliberation activation record fixed while varying the text observation boundary. (b) Repeated trajectories hold the source state and candidates fixed while sampled deliberations and realised actions vary; each trajectory is evaluated with activation and text readouts.

The matched-boundary diagnostic evaluates the stored trace at 0%, 25%, 50%, 75%, and 100% exposure. At each cutoff, the text observer receives that prefix and the activation reader is recaptured from the prompt ending at the same point. The five checkpoints form a coarse boundary sweep rather than a token-level timing study. The target remains the action produced after the original complete deliberation. Because the truncated prefixes did not themselves generate that action, this is a counterfactual matched-boundary diagnostic rather than a causal timing intervention.

Interpretation. At an intermediate cutoff, the truncated deliberation did not itself produce the target action. We recompute the activation from that prefix and ask how well both channels predict the action eventually generated after the complete stored deliberation. The sweep matches evidence boundaries for prediction; it does not claim that the model would have produced the same action had generation actually stopped at the cutoff.

Separately, the exposure/redaction series changes only the text supplied to position-balanced observers. Its 0%–100% labels share nominal cutoffs with the matched-boundary diagnostic but use a different evaluation aggregate and do not recapture activation; exact-order, decision-language, and mismatched-trace controls belong only to this series.

Residual streams are recorded from all 28 layers. The final prompt token, immediately before action generation, is the main activation feature. For temporal localisation we also use the last deliberation token and the mean over the deliberation span. Activation readers are standardised logistic regressions, with layer and regularisation strength $C \in \{0.01, 0.1, 1, 10\}$ selected inside each outer-training split. The small log-spaced C grid limits hyperparameter search on 40 source states.

Each prompted text observer is queried for the probability of candidate B. Qwen2.5-7B-Instruct supplies the same-family observer and Mistral-7B-Instruct-v0.3 a cross-family check. Queries are run in both candidate orders; probabilities are mapped back to a common orientation and averaged. We refer to this A/B-order averaging as *position balancing*. Appendix A.4 contains the prompt and parser details.

3.3 Scoring and validation

Our primary score is Brier loss,

$$\mathcal{L}_{\text{Brier}} = \frac{1}{n} \sum_{i=1}^{n} (p_i - y_i)^2. \tag{4}$$

Lower values are better. We use AUROC and within-state concordance when ranking is informative, and report accuracy, log loss, and calibration in the supplementary analyses. Within-state concordance compares opposite-action trajectory pairs from the same source state: it is the fraction for which the B trajectory receives a higher predicted probability of B than the A trajectory, with ties receiving half credit. Brier directly scores the probability assigned to the realised action, while AUROC and concordance isolate ranking behaviour.

For the 40-game analysis, paired uncertainty is estimated with a 20,000-replicate source-game bootstrap and 100,000 paired sign flips, supplemented by leave-one-game-out calculations. Activation capture ends before future action text. All 40 submitted fixed-action orders reproduce the stored transition when replayed with the native Diplomacy engine (native replay).

Cross-validation follows the same source structure. In the fixed-action experiment, layer and regularisation are selected from the outer-training data. In the repeated-trajectory follow-up, *complete-state holdout* means that all eight trajectories from a source state move together through inner selection and outer evaluation; no trajectory from a held-out state can enter fitting or model selection. A fixed grouped five-fold assignment (seed 20260825) supplies the primary point estimates, and 100 additional seeded assignments assess sensitivity to the partition. Its permutation null shuffles labels only within state, preserving each state's observed A/B count.

The fixed-action permutation control treats fold construction as part of its nested pipeline and therefore uses its own fixed cross-validation realisation. It reruns the real labels under that same procedure to obtain the 0.158 Brier reference for shuffled labels; this is not an alternative headline estimate to the primary 0.103 score. Appendix A.3 reports 100 seeded assignments to quantify partition sensitivity.

An exploratory fusion analysis asks whether activation adds value once matched full text is available. Text-only calibration and text-plus-activation fusion use nested logistic meta-models fit only on outer-training data; Appendix A.6 specifies the cross-fitting, regularisation selection, and results.

3.4 Repeated trajectories and text-reader sensitivities

The follow-up samples eight deliberation/action trajectories from each state at temperature 0.7 and top-p 0.9. Sampling creates within-state trajectory variation; the same settings are used for every state and are not themselves a target of inference. Among the 320 draws, 34 states produce both focal actions; these states contribute 434 opposite-action trajectory pairs with an identical board position and candidate set. Every trajectory has its own generated deliberation and realised action. Native replay succeeds for all 320 submissions.

We evaluate the same three activation summaries used for temporal localisation. For text, the original comparator is a greedy Qwen observer prompted for a probability. Three sensitivity analyses change the readout while leaving the underlying trajectory fixed. Direct same-family target scoring does not ask Qwen to generate a numerical probability: it scores fixed YES and NO completions for the candidate-choice query under the causal language model and normalises those likelihoods. The other readers are a cross-family greedy Mistral observer and a supervised character 3–5-gram TF–IDF logistic model trained only on complete training states. The supervised reader receives task labels and no activation features.

Disclosure is measured separately. Automated lexical checks record exact candidate mentions over all 320 trajectories. A randomised packet of 80 traces – two per source state – was scored with the action key withheld using a fixed selection-language rubric that marks explicit recommendations or commitments. The rubric is deterministic and is reported as a rule-based disclosure diagnostic. Full follow-up details are in Appendix A.7.

4 Fixed-action results

4.1 Observation-boundary reversal

On the primary five-fold out-of-fold predictions, the post-deliberation activation reader scores 0.103 Brier. The early text observer is at 0.231; after receiving the generated deliberation it falls to 0.033. Thus $\Delta_0 = +0.128$ and $\Delta_1 = -0.069$: with L_A fixed at 0.103, only text exposure changes the sign. At the two matched endpoints, text also has lower Brier than activation both before deliberation (0.231 versus 0.252) and after it (0.033 versus 0.103). The reversal is specifically the change in activation-versus-text ordering; it does not depend on activation significantly outperforming the empirical-prior baseline discussed in Section 4.3. Table 1 places the matched before/after scores beside the paired contrasts. The boundary sweeps below use different aggregates, so their nominal endpoints are not alternative estimates of these values.

Table 1: **Matched text is lower-Brier at both fixed-action observation points.** The lower block reports text-minus-activation contrasts relative to the post-deliberation activation reader; positive values favour activation. Contrasts use full-precision predictions before display rounding.

Observation point	Activation Brier	Text Brier
Before deliberation	0.252	0.231
After deliberation	0.103	**0.033**
Contrasts to post-deliberation activation (0.103)		
Comparator	Comparator Brier	Difference (95% CI)
Early text	0.231	$+0.128\ [0.010, 0.243]$
Fold-trained empirical prior	0.195	$+0.092\ [-0.020, 0.200]$
Full-deliberation text	0.033	$-0.069\ [-0.147, -0.001]$

4.2 What the deliberation reveals

Every Qwen deliberation contains the selected candidate verbatim; 39 of 40 contain both candidates, and the last exact candidate mention agrees with the submitted action in all 40 cases. Position-balanced Qwen and Mistral observers move from 0.208 to 0.048 and from 0.212 to 0.007 Brier, respectively, when the full trace is supplied. The near-verbatim choice language gives them direct predictive evidence about the eventual action, but does not show that the verbalised reasoning caused that action. Figure 2a includes the redaction controls.

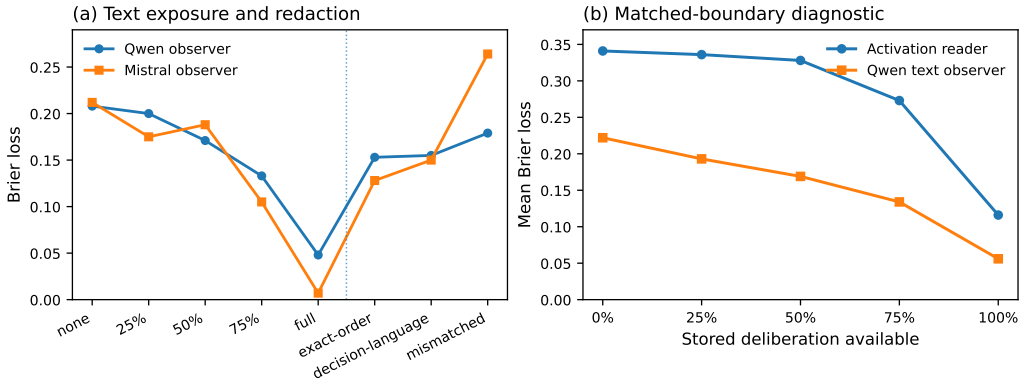


Figure 2: **Text improves with deliberation exposure and remains lower-Brier at every tested matched boundary.** (a) The exposure/redaction series varies text supplied to position-balanced Qwen and Mistral observers. (b) The matched-boundary diagnostic recaptures both activation and text and averages 20 grouped assignments; text has lower Brier in all 20 assignments at every boundary. The panels and Table 1 use different evaluation aggregates.

Figure 2 juxtaposes the two designs: panel (a) changes only observer text, whereas panel (b) recaptures both channels. Shared percentages are labels within each analysis, not a common set of point estimates.

The exact-order mask removes 147 spans, about 1.5% of characters. The broader decision-language rule removes 199 spans and about 18% of characters. Prediction worsens under both masks, although comparative and preference language survives in some traces. Appendix A.5 describes the masks; Appendix A.6 contains the full condition table.

For the matched-boundary diagnostic we retain the action from the complete stored trajectory and recapture both monitor inputs at 0%, 25%, 50%, 75%, and 100% of the deliberation. As Figure 2b shows, text has lower mean Brier at every boundary and wins all 20 grouped assignments at each one. Appendix A.6 reports the numerical means.

4.3 Controls and extensions

The remaining checks probe sensitivity to the baseline, reader-selection procedure, channel fusion, and model family or scale.

The prevalence comparison is a baseline check rather than part of the reversal. The 10/30 action split makes a constant, fold-trained prevalence predictor fairly competitive at 0.195 Brier. Its paired difference from the post-deliberation activation reader is 0.092, with bootstrap interval $[-0.020, 0.200]$ and sign-flip $p = 0.113$.

The selection-aware permutation analysis treats fold construction as part of the null pipeline: it rebuilds the five folds with a fixed seed and repeats layer/C selection for every target. Its real-label reference is therefore 0.158 Brier under that self-contained procedure, rather than the 0.103 point estimate from the primary evaluation split. One of 500 shuffled tasks scores below 0.158 ($p = 0.004$); Appendix A.3 reports the broader 100-assignment fold-sensitivity analysis.

An exploratory fusion analysis averages 0.020 Brier for calibrated text plus activation versus 0.026 for calibrated text alone; fourteen of 20 assignments improve, with changes from -0.023 to $+0.024$ (Appendix A.6).

The model-family extensions use the same 40 states. A fresh Mistral subject agrees with the Qwen subject on 28 states, and both Mistral and Qwen observers improve when they see its full deliberation. At 32B scale, OLMo-2 observing Qwen3 moves from 0.324 to 0.193 Brier; Qwen3 observing OLMo-2 moves from 0.213 to 0.053. Appendix A.6 contains the remaining replication results, including the non-monotone Mistral exposure series.

5 Repeated trajectories and reader sensitivity

The fixed-action experiment has one outcome per source position, so it cannot distinguish state-level propensity from trajectory-specific information. We therefore draw eight trajectories from each of the same 40 states and score each against its own action. Thirty-four states produce both focal actions, yielding 434 opposite-action pairs that provide the comparison unavailable in the fixed-action sample: different realised actions from the same strategic position and candidate set. All fitting and model selection use complete-state holdout. Because this supervision differs from the fixed-action regime, the repeated-trajectory score is not simply a larger-sample estimate of the 0.103 fixed-action score.

With the final prompt token as the representation, every held-out trajectory in the primary grouped cross-validation assignment is classified correctly. The activation is captured before any action text is generated. On this assignment, Brier is 3.2×10^{-7} and AUROC and within-state concordance are both 1.0. Earlier summaries remain strong – 0.031 Brier at the last deliberation token and 0.046 for the mean deliberation span – but probabilistic loss drops sharply close to action generation. Across 100 alternative grouped fold assignments, the final-prompt reader keeps perfect accuracy, AUROC, and concordance and averages 1.5×10^{-6} Brier. Selected layers cluster near the end of the network, principally 24–26; Appendix A.7 gives details.

The near-zero loss naturally raises a leakage concern. The action token itself is not part of the captured activation, and complete-state holdout excludes every trajectory from a held-out source state from fitting and model selection. These safeguards rule out direct inclusion of the generated action and cross-state train/test contamination; they do not rule out very late output preparation, which remains one interpretation of the result.

The permutation test keeps the A/B count of each source state fixed and shuffles labels only among its eight trajectories. Reader selection is rerun for every shuffled target. Over 500 draws, null Brier ranges from 0.231 to 0.358; none approaches the observed score ($p = 0.002$). The within-state-concordance permutation value is also $p = 0.002$.

The text side changes much more when the readout changes. Greedy Qwen scores 0.044 Brier after seeing the trajectory-specific deliberation. Direct Qwen target scoring, computed from fixed YES/NO likelihoods on the same prompts, reaches 0.008 Brier with AUROC 0.995 and within-state concordance 0.991. The Mistral observer reaches 0.018 after deliberation, compared with 0.256 before it. A supervised character 3–5-gram TF–IDF logistic model, despite receiving task labels, performs poorly at 0.205 on the primary assignment and 0.204 on average across 20 complete-state-held-out assignments.

Table 2: **Text performance changes sharply with the readout used.** Fitted readers use complete-state holdout, while prompted and direct language-model readouts require no task fitting; secondary metrics and split-sensitivity results appear in Appendix A.7.

Readout	Brier
Final prompt activation	3.2×10^{-7}
Greedy Qwen full text	0.044
Direct Qwen target scoring	0.008
Greedy Mistral full text	0.018
Supervised character 3–5-gram text	0.205

The comparison is metric-dependent: activation versus direct Qwen is 1.0 versus 0.995 AUROC and 1.0 versus 0.991 within-state concordance, while greedy Qwen is at 0.982 and 0.974. For the greedy-Qwen comparison, the state-paired full-text-minus-activation squared-error difference is 0.044 with source-state bootstrap interval $[0.034, 0.055]$, and the sign survives every leave-one-state-out calculation.

Candidate strings are present almost everywhere in the trajectory text. At least one exact candidate string appears in 319 of 320 traces, both strings appear in 315, and the last exact mention matches the realised action in 302. The action-key-withheld disclosure packet contains 80 traces; a fixed selection-language rubric marks an explicit recommendation or commitment in 65. The trajectory text therefore contains target-relevant evidence, leaving extractability under the reader as the unresolved issue. Since the final-prompt activation is measured only after this deliberation, the near-zero Brier primarily establishes exceptionally high late accessibility. Where else the target information can be recovered must be assessed from the text readouts themselves. Detailed diagnostics are in Appendix A.7.

6 Discussion

The fixed-action comparison makes the timing problem concrete. Once the text observer receives the deliberation preceding the activation record, the apparent activation advantage reverses, despite the activation reader passing a selection-aware shuffled-label control. The matched-boundary diagnostic extends the ordering beyond the full-trace endpoint, although five cutoffs cannot localise a crossover at token resolution. Comparing mismatched observation points can therefore mix channel access with later trajectory evidence.

Direct target scoring exposes a second source of sensitivity. It reduces Qwen's text Brier from 0.044 to 0.008, with AUROC 0.995 and concordance 0.991 against 1.0 for activation. Activation remains numerically best among the tested readers, yet the ranking gap is small under this readout. Some of the residual difference may reflect unusually easy extraction from the late residual stream. Earlier availability, lower cost, or robustness to evasion could still give activations operational value; this benchmark does not measure those properties.

The repeated trajectories clarify what the late activation result itself establishes. Under complete-state holdout, the final-prompt residual state carries highly accessible trajectory-specific information, with the strongest signal immediately before action generation. The deliberation is also rich in action-relevant evidence. This temporal pattern fits either final output preparation or an earlier commitment that remains readable into the last prompt state, and the present measurements do not separate them.

These comparisons also inherit asymmetries from the task itself. Text receives named candidates and preference language, whereas activation is measured immediately before a constrained action. The benchmark is therefore a test of sensitivity to comparison design, not a general ranking of monitoring channels.

Distinguishing output preparation from earlier commitment requires denser temporal measurement. A sharp rise near the action boundary would favour the former; a useful signal before the text becomes explicit would support the latter. Interventions on the decoded direction could then test causal involvement [5].

7 Conclusion

The experiments identify two distinct sources of instability in comparisons between activation and text monitoring. Matching the information available at the observation boundary reverses the fixed-action comparison. In repeated trajectories, the late residual state tracks which action is realised within the same strategic position. Direct target scoring on the text sharply reduces the apparent margin. The strongest empirical claim concerns late internal accessibility; the ordering of monitoring channels is conditional on the comparison design.

That distinction changes how monitoring results should be reported. Observation timing belongs in the experimental specification because later traces can contain evidence unavailable to an earlier baseline. Reader construction deserves the same status when one channel can be probed in several plausible ways. A channel comparison is consequently meaningful only relative to the evidence boundary and extraction procedure used for each side.

Temporal localisation is now the most direct follow-up. Denser activation capture through deliberation could reveal whether action-predictive structure precedes explicit textual commitment or rises mainly during output preparation. If an early internal signal survives stronger, more closely matched text readouts, that would support a more operationally significant activation advantage. Causal intervention on such a signal would then test whether it participates in the decision process rather than merely recording a choice already formed elsewhere.

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A Supplementary material

A.1 Why Diplomacy

Negotiation, social-dilemma, and hidden-role benchmarks provide useful precedents for studying strategic model behaviour [1, 20, 11]. Their observations are often clustered within conversations, repeated games, or shared hidden states, which makes the unit of analysis as important as the row count. Diplomacy is convenient for the present question because legal orders are mechanically checkable and released CICERO games provide identifiable source situations [12]. We use one movement-phase state per source game for the fixed-action evaluation and reuse those states only for the explicitly grouped stochastic experiment.

A.2 State construction and subject generation

The evaluation bank is constructed from released CICERO gameplay records, with one movement-phase state selected from each of 40 distinct source games before any local Qwen or Mistral outcome is generated. Candidate phases must contain at least two released messages involving the focal power. Within each game, candidates are considered deterministically in decreasing conversation volume, then by phase and power, with at most 18 candidates examined. The native CICERO engine reconstructs the released parent state and enumerates legal focal-order pairs while all non-focal source-game replay orders remain fixed. A pair is admissible only when both alternatives are legal under that background and produce different post-phase board states. Admissible pairs are ranked by a fixed rule favouring distinct order types and mentions of order-relevant locations in the released messages; deterministic lexical tie-breaking selects the top pair. A SHA-256-based swap then assigns the two orders to A/B. Thus the 40 states are protocol-selected from released games rather than randomly sampled or selected in response to local-model behaviour.

Each record contains the model-visible board state and released bilateral messages involving the focal power, fixed own non-focal orders, and the two focal candidate orders. The Qwen subject receives this state and an instruction to analyse the two focal orders in a delimited deliberation field. A subsequent action prompt presents the same state and generated deliberation and requests exactly one focal order in a separate field, which is withheld from every text observer.

Greedy generation produces 10 A and 30 B choices. All 40 submitted actions pass native replay. Observer exposure and redaction conditions reuse the same parent states, candidate pairs, deliberations, and actions.

A.3 Activation capture and reader fitting

The fixed-action activation reader is built from residual streams saved at all 28 layers of the 7B Qwen subject. The main representation is the final prompt token after deliberation and immediately before action generation. The main reader search uses that final-prompt representation; the last deliberation token and mean deliberation span are evaluated separately for temporal comparison.

Within each outer training split, activations are standardised before logistic regression and $C \in \{0.01, 0.1, 1, 10\}$ is selected on training data only. The 40-state analysis uses five outer folds, and 100 seeded fold assignments measure sensitivity to the partition. As a sensitivity analysis, reader selection may also choose among all three temporal summaries. This broader search does not improve mean Brier: the final-prompt representation is selected in 453 of 500 outer folds, and the broader search is slightly worse on average.

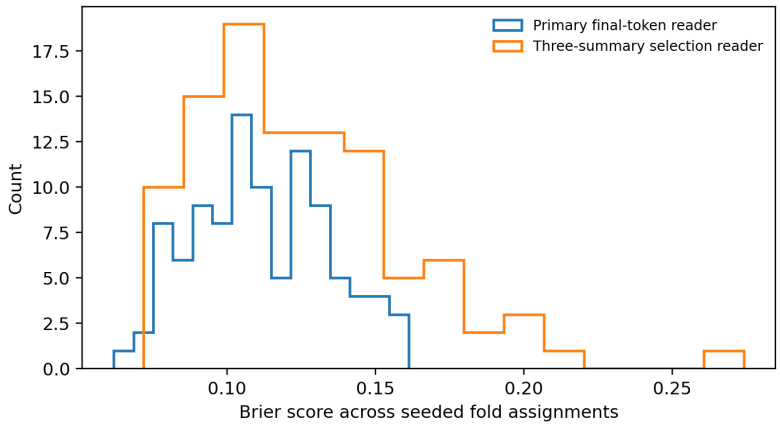


Figure 3: **Searching over temporal summaries does not improve mean activation-reader performance.** Brier loss across 100 five-fold assignments in the Qwen experiment; the broader search increases variability.

A.4 Observer prompts and position balancing

Each prompted observer sees the same rendered state as the subject and reports the probability that candidate B will be chosen. Every query is issued twice with A/B order reversed; probabilities are mapped back to a common orientation and averaged. We call this averaging position balancing. Qwen2.5-7B-Instruct is the same-family observer and Mistral-7B-Instruct-v0.3 supplies the cross-family comparison.

Probability extraction accepts only unambiguous numeric forms. Of 640 Mistral-observer outputs in the exposure/redaction analysis, 581 are bare numbers; the remaining valid outputs use a small set of explicitly specified numeric formats. Numeric intervals are mapped to their midpoint, and free-form prose is never interpreted as a probability.

A.5 Exposure and redaction construction

Exposure-series conditions truncate the stored deliberation at tokenizer-aligned 25%, 50%, and 75% boundaries. The exact-order mask removes known textual variants of the candidate orders and option labels. The decision-language mask additionally removes sentences containing a fixed list of decision verbs.

The exact mask removes 147 spans, averaging 1.5% of characters. The decision-language rule removes 199 spans, including 107 decision sentences, and averages 18.0% of characters. Comparative or conclusion-like cues remain in some decision-language-redacted traces. The redaction conditions therefore measure lexical removal, with semantic preference cues still present. The stochastic traces are similarly explicit: 319 of 320 contain an exact candidate string, and in 302 of 320 the last exact candidate mention matches the chosen action. A separate generation instruction intended to avoid

naming either focal order also failed to suppress such mentions reliably, and no reader comparison from that condition is used.

For the mismatched-trace control, decision-language-redacted traces are permuted across other source games within deliberation-length quartiles. This keeps rough length and formatting similar while breaking the original state–trace pairing.

A.6 Supplementary fixed-action analyses

Observer metrics. Brier loss is the primary score, but the exposure effect also appears in log loss and accuracy. From no deliberation to full deliberation, Qwen observer log loss falls from 0.614 to 0.228 and accuracy rises from 0.675 to 0.975. Mistral moves from 0.674 to 0.069 log loss and from 0.750 to 1.000 accuracy. Both full-text AUROCs are 1.0. Five-bin expected calibration error (ECE) changes from 0.151 to 0.159 for Qwen and from 0.133 to 0.065 for Mistral.

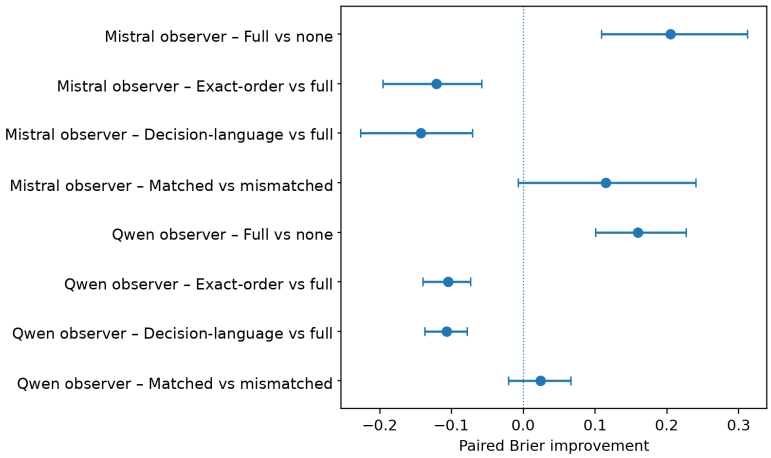


Figure 4: **Paired source-game contrasts quantify the exposure/redaction effects.** Brier improvements in the Qwen-subject experiment are shown with 95% bootstrap intervals; positive values indicate improvement of the named condition over its comparison condition.

Table 3: **Both observers improve substantially with full deliberation for the Qwen subject.** Position-balanced Brier loss by exposure/redaction condition.

Condition	Qwen observer	Mistral observer
No deliberation	0.208	0.212
25% exposure	0.200	0.175
50% exposure	0.171	0.188
75% exposure	0.133	0.105
Full deliberation	**0.048**	**0.007**
Exact-order redaction	0.153	0.128
Decision-language redaction	0.155	0.150
Mismatched redacted trace	0.179	0.264

Prevalence baseline. With 30 of 40 Qwen actions labelled B, the fold-trained constant predictor has overall Brier 0.195. The post-deliberation activation reader improves on it by 0.092; the bootstrap interval is $[-0.020, 0.200]$, the paired sign-flip value is $p = 0.113$, and leaving out any one source game preserves the sign of the difference.

Permutation test with nested reader selection. The fully nested permutation control treats fold construction as part of its pipeline and therefore uses a real-label reference Brier of 0.158 under its own fold construction; Section 4.3 explains why this differs from the primary 0.103 point estimate.

Across 500 shuffled targets, mean Brier is 0.331 (median 0.333; range 0.155–0.522), with one shuffled task below the real-label reference ($p = 0.004$).

Fresh Mistral subject. The valid Mistral run yields 16 A and 24 B choices, agrees with Qwen on 28 states, and passes native replay in all 40. Both the same-family Mistral observer and cross-family Qwen observer improve when given the complete Mistral deliberation, and both are weaker after exact-order or decision-language redaction.

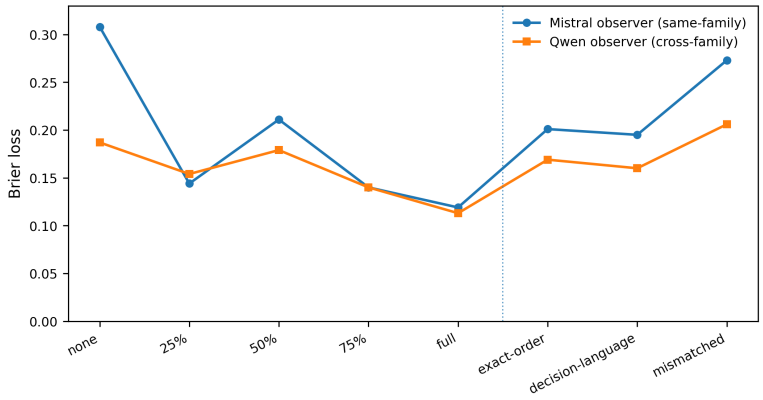


Figure 5: **Full deliberation improves both observer families for the Mistral subject.** Exact-order and decision-language redaction worsen their predictions.

Table 4: **The Mistral-subject replication reproduces the full-deliberation improvement.** Position-balanced Brier loss by exposure/redaction condition.

Condition	Mistral observer	Qwen observer
No deliberation	0.308	0.187
25% exposure	0.144	0.154
50% exposure	0.211	0.179
75% exposure	0.140	0.140
Full deliberation	**0.119**	**0.113**
Exact-order redaction	0.201	0.169
Decision-language redaction	0.195	0.160
Mismatched redacted trace	0.273	0.206

Across 100 grouped assignments, the Mistral activation reader averages 0.153 Brier. Full text is lower by 0.034 for the same-family observer (bootstrap interval $[-0.039, 0.108]$, $p = 0.389$) and by 0.040 for the cross-family Qwen observer ($[-0.023, 0.107]$, $p = 0.244$).

Matched-boundary diagnostic and fusion. The fixed-target matched-boundary diagnostic recaptures activations from progressively longer versions of the original trace while keeping the final action unchanged. The truncated prefixes did not themselves generate that action, so this is a counterfactual matched-boundary diagnostic rather than a causal timing intervention.

Table 5: **Text has lower Brier in every tested matched-boundary assignment.** Means are over 20 grouped reader assignments; the text column is the Qwen text observer, and the final column counts assignments in which activation has lower Brier than text.

Deliberation available	Activation mean	Text Brier	Activation wins
0%	0.341	0.222	0/20
25%	0.336	0.193	0/20
50%	0.328	0.169	0/20
75%	0.273	0.134	0/20
100%	0.116	0.056	0/20

The fusion analysis is exploratory. In each of 20 seeded five-fold assignments, outer-training activation probabilities are generated by four-fold cross-fitting, with representation, layer, and activation-reader C reselected inside each cross-fit split; outer-test probabilities use a reader selected and fitted on the full outer-training fold. Text and activation probabilities are clipped to $[10^{-5}, 1 - 10^{-5}]$ and transformed to logits. A standardised `liblinear` logistic meta-model is fit either to the text logit alone (calibration) or to text and cross-fitted activation logits (fusion), with $C \in \{0.01, 0.1, 1, 10, 100\}$ selected by four-fold stratified cross-validation inside the outer-training data. Outer-test rows remain untouched until evaluation.

Across the 20 assignments, activation alone averages 0.116 Brier, raw full text 0.033, calibrated text 0.026, and text plus activation 0.020. Relative to calibrated text, the average gain is 0.006; fourteen assignments improve and six worsen, with differences from -0.023 to $+0.024$.

32B cross-family extension. This extension keeps the same 40 states while changing scale and family on both sides of the observer comparison; Qwen3 and OLMo-2 each pass 40/40 native replay. With Qwen3 as subject, OLMo-2 moves from 0.324 to 0.193 Brier after seeing the deliberation, a 0.131 reduction with bootstrap interval $[0.073, 0.194]$. Reversing the roles, Qwen3 observing OLMo-2 moves from 0.213 to 0.053, a 0.160 reduction with interval $[0.119, 0.208]$. Decision-language redaction worsens both directions; for Qwen3, the activation reader averages 0.037 Brier across 100 assignments, compared with 0.397 for the shuffled-label control applied to the same Qwen3 activation-reader analysis.

A.7 Repeated stochastic-trajectory follow-up

Each of the 40 source states is sampled eight times at temperature 0.7 and top-p 0.9. The resulting 320 trajectories contain 126 A and 194 B actions. Thirty-four states produce both actions, yielding 434 opposite-action pairs with the same board position and candidate orders. All 320 submitted actions pass native replay from their released parent states. Reader fitting is grouped by state ID; every trajectory from a held-out state is excluded from inner selection and outer fitting.

Grouped activation analysis. The final prompt token is fixed as the primary representation. In the primary grouped cross-validation assignment its Brier loss is 3.2×10^{-7}, with accuracy, AUROC, and within-state concordance all equal to 1.0. Brier is 0.031 at the last deliberation token and 0.046 for the mean deliberation span. Across 100 alternative grouped fold assignments, the final-prompt reader remains at 1.0 on accuracy, AUROC, and within-state concordance and averages 1.5×10^{-6} Brier. Selected layers cluster near the end of the network, principally layers 24–26.

Table 6: **The final-prompt activation readout is strongest, while full-text ranking is already high.** Metrics are for the primary grouped cross-validation assignment. Within-state concordance compares opposite-action trajectories from the same source state, with ties receiving half credit.

Predictor	Brier	AUROC	Within-state concordance
Final prompt activation	3.2×10^{-7}	1.000	1.000
Last deliberation token	0.031	0.993	0.993
Mean deliberation span	0.046	0.988	0.998
Greedy Qwen full text	0.044	0.982	0.974

State-paired full-text minus activation squared error is 0.044 with source-state bootstrap interval $[0.034, 0.055]$; leaving out any one state preserves its sign. The nested null shuffles labels among the eight trajectories of each state, keeping the observed A/B count fixed and rerunning layer/C selection. Across 500 draws, null Brier averages 0.292 (SD 0.022; range 0.231–0.358). The observed activation score lies below all 500 draws ($p = 0.002$); the within-state-concordance permutation value is also $p = 0.002$.

Text reader sensitivity. The original greedy Qwen full-text observer has Brier 0.044, accuracy 0.988, AUROC 0.982, and tie-adjusted within-state concordance 0.974. Direct same-family target scoring uses the stored full-deliberation prompts and fixed YES/NO completions rather than a generated numerical probability; normalised causal-LM likelihoods produce the target probability. It reaches 0.008 Brier, 0.995 AUROC, and 0.991 within-state concordance. The Mistral-7B-Instruct-v0.3 observer scores 0.018 Brier after deliberation and 0.256 before it. The supervised TF–IDF logistic reader uses character 3–5-grams within word boundaries; vocabulary, IDF, regularisation selection, and fitting are restricted to complete training states. It scores 0.205 Brier on the primary assignment and averages 0.204 across 20 grouped assignments.

Disclosure diagnostics. Exact candidate strings occur in 319 of 320 trajectories; both strings occur in 315, and the last exact candidate mention matches the realised action in 302. The randomised 80-trace packet contains two trajectories per source state and withholds the action key during scoring. A fixed selection-language rubric marks explicit recommendation or commitment language in 65 traces (81.25%). The scorer is deterministic, so this percentage is a rule-based diagnostic of explicit choice language.

A.8 Limitations

The task reduces each decision to two named legal orders with the focal power's remaining orders fixed, and all family/scale extensions reuse the same 40 source games. This setup favours late trajectory prediction: candidate names appear in the prompt and often in the deliberation, while the strongest activation is taken immediately before a constrained action field. Wider action sets, unnamed candidates, genuinely concealed deliberation, or a new state bank could change both the activation result and the usefulness of direct target scoring. The Mistral and 32B extensions are robustness checks on this bank, not independent task or dataset replications.

The repeated-trajectory follow-up has 320 trajectories but only 40 independent strategic positions. Grouped fitting, source-state bootstrap intervals, and leave-one-state-out calculations preserve that unit, so uncertainty remains conditional on the selected bank. The states are protocol-selected from released CICERO games rather than probability-sampled from a defined population. In the fixed-action experiment, the activation reader's improvement over the fold-trained empirical prior has a bootstrap interval including zero and a paired sign-flip value of $p = 0.113$. A second independently constructed state bank would address external validity more directly than additional trajectories from the same positions.

The readers do not have matched capacity or cost: the activation reader, prompted probability observers, direct language-model score, and supervised character model differ in representation, supervision, and inference-time compute. On repeated trajectories, changing only the text readout moves Brier from 0.044 to 0.008. None of the tested text readers matches the final activation reader, but the tested set does not identify an optimal reader for either channel. We therefore do not estimate a compute- or supervision-matched frontier.

Five exposure points cannot determine how many tokens or how much wall-clock time earlier one channel becomes useful. The matched-boundary ordering holds only at those cutoffs, leaving a possible early-warning advantage for internal monitoring unresolved. No activation was patched, removed, or steered, so the decoded direction's causal role is open [5]. Monitor-aware or adversarial generation was not tested; visible reasoning could become less informative in that setting [9].

Measurement is also limited. Brier is an overall probabilistic loss rather than a pure calibration measure, and five-bin calibration summaries on 40 fixed-action states are necessarily descriptive. The 80-trace disclosure packet uses a deterministic selection-language rubric rather than independent human semantic judgments. Finally, two 500-draw permutation analyses resolve their extreme tails only coarsely, so values near 0.002–0.004 should be read at that Monte Carlo resolution.

A.9 Compute and data availability

Repeated-trajectory and Qwen/Mistral reader runs used NVIDIA A100 80GB PCIe GPUs; the 32B extension used an A100 pod. Incomplete timing records preclude aggregate GPU-hour reporting. Analysis artifacts from the private double-blind repository are mirrored in an anonymous archive at `https://osf.io/tzr9n/overview?view_only=8b55bb057654432088dd120df0395ea8`, subject to the original CICERO distribution terms.

Ethics and dual use. The study uses released game states and model-generated traces, not human-subject data. Pre-action monitoring requires internal telemetry and could itself become a training target; adaptive evasion and telemetry-privacy risks are outside the present experiments.